

KnackELT Architecture

How knack-elt works, and where it sits in a full Knack → warehouse → BI stack.

Four diagrams:

1. [Reference architecture](#) — the end-to-end stack, in [plain language](#) and with [the technical detail](#)
2. [Pipeline flow](#) — what knack-elt actually does on a run
3. [Run sequence](#) — the call order for one invocation
4. [SCD2 row lifecycle](#) — how history is represented, and how to query it

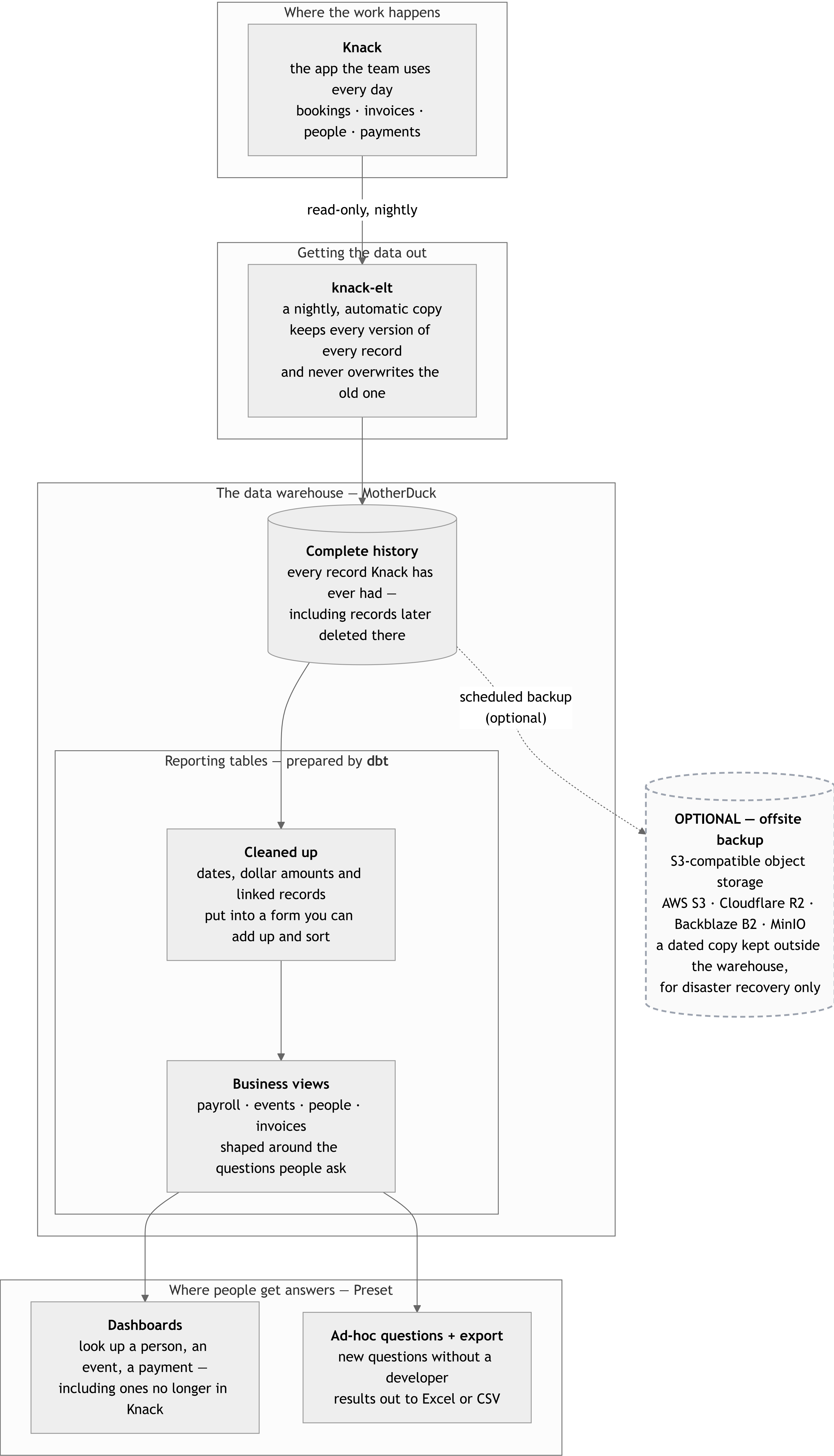
Scope note. This repo is the **ingestion** piece only. The dbt models, BI configuration, and CI orchestration shown in diagram 1 are drawn from a working production deployment where they live in a companion application repo (whose `pipelines/knack_dlt.py` is the production twin of this repo's pipeline). They are real, not aspirational — but they are not in this tree. Subgraph labels mark what lives where.

1. Reference architecture

Knack is the system of record and the operational UI. It is not a warehouse: no SQL, no aggregates, no history, and a record cap that eventually forces pruning. The stack below turns it into one — MotherDuck as the warehouse, dbt as the transformation layer, Preset (managed Apache Superset) as the access layer.

Same architecture, drawn twice: once for a stakeholder audience, once with the detail an engineer needs.

The stack in plain language

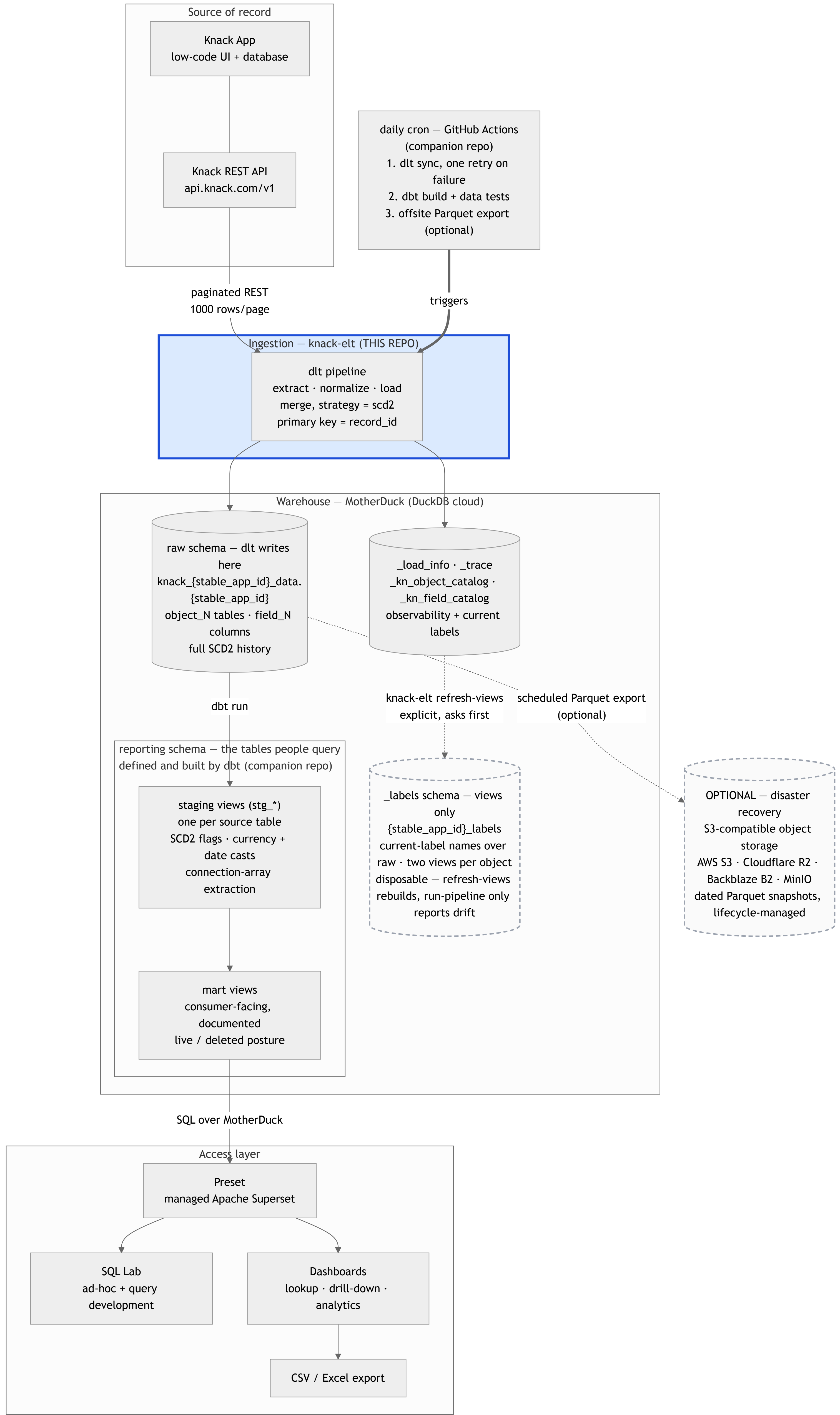


How to read it. The copy runs one way only — nothing this stack does can change data in Knack. The warehouse keeps history, so a record deleted in Knack is still answerable afterwards; that is the whole point of having it. The two boxes inside the warehouse are a division of labour, not two copies of the data: the complete history is what arrives, and **dbt** is the tool that turns it into the cleaned, business-shaped tables people actually query.

The dashed box is genuinely optional. The warehouse is already a second copy of Knack, so this is a third — insurance against losing the *warehouse* (a bad migration, an account problem, a provider outage), not against losing Knack. Add it when the warehouse becomes the only remaining copy of pruned records; skip it while Knack still holds everything. Any S3-compatible store works, and retention is normally handled by the bucket's own lifecycle rules rather than by code.

The same stack, with the technical detail

FIGURE 2 — THE SAME STACK, WITH THE TECHNICAL DETAIL



The load-bearing detail: schema ownership is split.

Schema	Owner	Contents	Rule
raw (the stable app-id dataset)	dlt	One <code>object_N</code> table per Knack object, <code>field_N</code> columns, full SCD2 history	dbt never builds into it. It is the pipeline's output, and a <code>dbt run</code> writing here would be clobbered on the next sync.
<code>{stable_app_id}_labels</code>	knack-elt (<code>labels.py</code>)	Two views per object — <code>"Customers"</code> (live rows) and <code>"Classes_history"</code> (every version, plus <code>valid_from</code> / <code>valid_to</code> / <code>is_live_in_knack</code>)	Views only, disposable, rebuilt wholesale by <code>knack-elt refresh-views</code> — never automatically, never touching <code>object_N</code> / <code>field_N</code> .
<code>reporting</code>	dbt	Staging + mart views	Everything downstream reads here. No dashboard queries raw directly.

Label views are this repo's, not the companion repo's. `{stable_app_id}_labels` is generated from `_kn_object_catalog` / `_kn_field_catalog` — the labels loaded on every sync — so an analyst can browse `"Customers"` instead of `object_3` . It is disposable and read-only: identifier comparisons are ASCII-case folded so `"Customers"` and `"customers"` can't collide, but every name emitted into SQL is verbatim, and an apply is a full drop-and-rebuild in one transaction rather than an incremental diff, because an incremental create-then-drop can transiently clobber a view if two objects swap the names their views hold. Nothing rebuilds this layer on its own — a label edited in the Knack builder must never move a warehouse name by itself. `knack-elt refresh-views` prints a plan and asks before applying it; `run-pipeline` only reports drift at the end of a sync and changes nothing. A hand-authored view in the schema is removed on the next apply; a hand-authored table blocks the rebuild instead and is named in the error.

Staging exists so that exactly one layer absorbs the mess Knack and dlt produce together — currency strings with commas, dates wrapped in JSON, connection fields as JSON arrays, and dlt's type-variant columns (when a value doesn't fit the type inferred from the first batch, dlt NULLs the base column and diverts the value to a sibling like `amount__v_double` , which makes a naive `SUM()` under-report silently). Marts and dashboards then get to be simple.

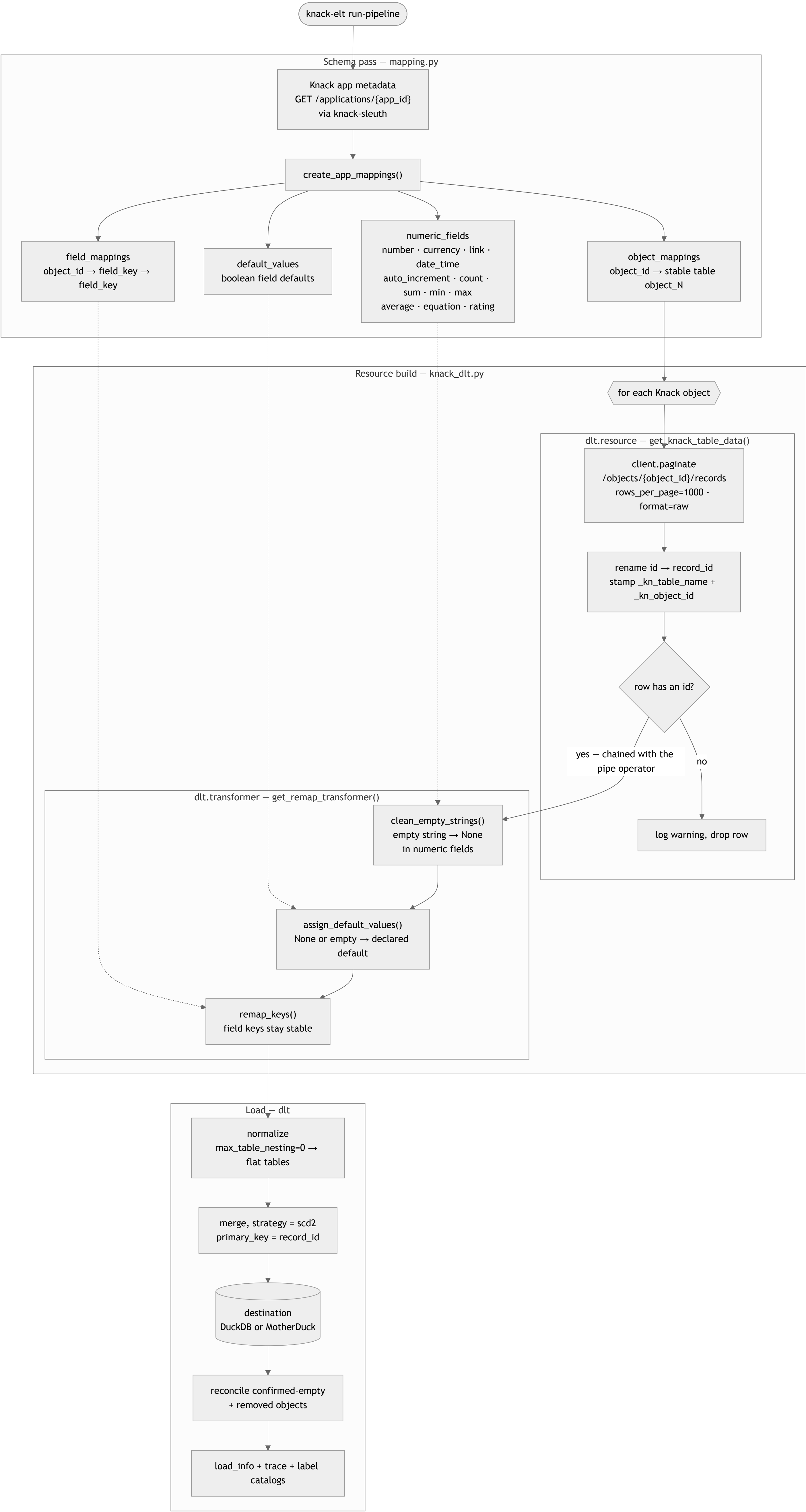
Why Preset: it is managed Superset, so SQL Lab doubles as the query-development environment and the dashboard builder against the same MotherDuck connection — a proven query becomes a dataset becomes a chart with no rewrite. Any BI tool with a DuckDB/MotherDuck SQLAlchemy connection

substitutes here; the contract is the `reporting` schema, not the tool. Note that row-level security and embedded analytics are paid-tier features in Preset — untrusted multi-tenant self-service needs a different surface (a scoped API in front of MotherDuck), not a BI seat.

2. Pipeline flow

What one run of knack-elt does. Two inputs from Knack: the **app metadata** (schema) and the **records** (data). Immutable keys identify physical tables and columns; metadata labels are loaded into catalog tables for discovery.

FIGURE 3 — PIPELINE FLOW



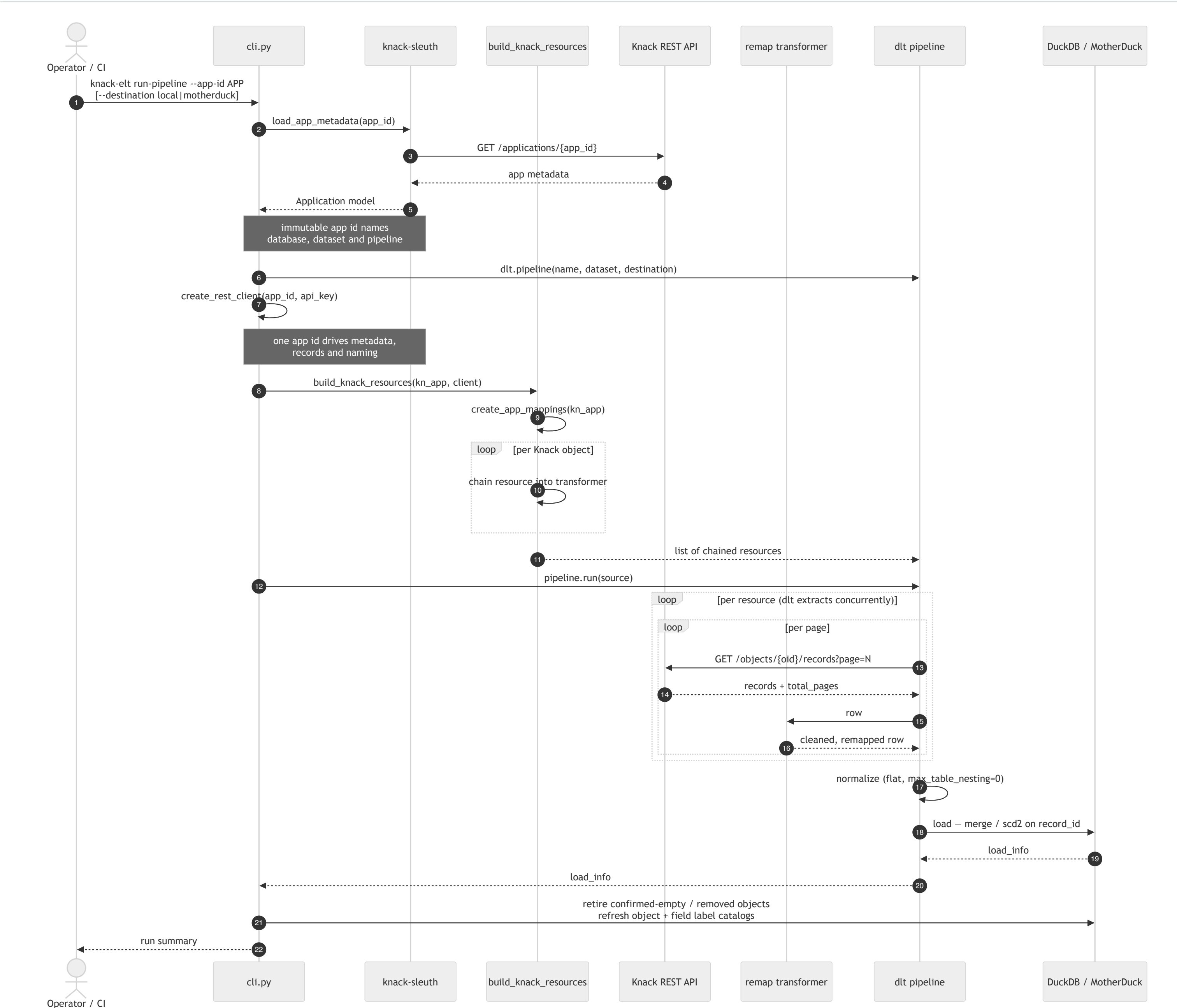
Notes on the mapping pass

- Object and field labels are editable metadata, so they never identify physical warehouse objects. `object_12` remains table `object_12`; `field_73` remains column `field_73`.
 - `_kn_object_catalog` and `_kn_field_catalog` map those stable keys to current labels for discovery and downstream model generation.
 - Label changes, duplicate/non-Latin labels, and names matching top-level system keys therefore cannot split history or overwrite a value.
 - Knack's row id arrives as the top-level `id` key and is renamed to `record_id`, which is the pipeline's primary key. A field labelled "ID" remains under its immutable `field_N` column. The key is taken from the payload, not from Knack's auto-added "Record ID" field, so it works identically on apps that predate that field.
 - Cleaning runs **before** the remap only, so it matches on raw Knack field keys. `numeric_fields` and `default_values` are registered under those raw keys only.
 - `remap_keys` falls back to the original key when no mapping exists, so an unmapped object still loads — with `field_NN` column names.
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3. Run sequence

The packaged entry point is the Typer CLI (`knack-elt run-pipeline`). It takes the app metadata from `knack-sleuth` rather than fetching it itself, and builds the record client from the *same* app id and key, so metadata and records always describe one app.

FIGURE 4 – RUN SEQUENCE



One entry point, two destinations:

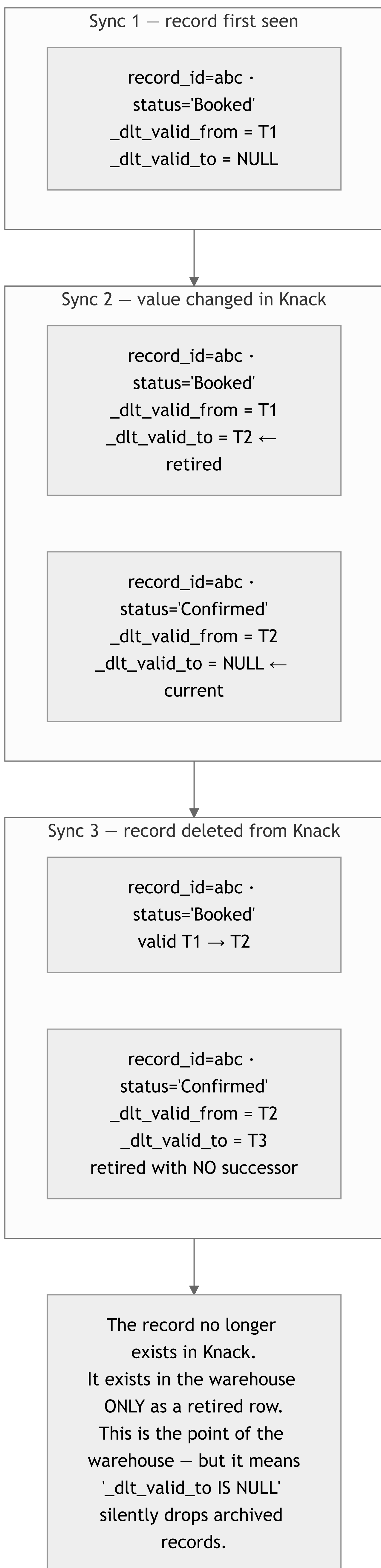
-- destination	Target	Requires
<code>local</code> (default)	a DuckDB file at <code>\$XDG_DATA_HOME/knack-elt/knack_{stable_app_id}_data.duckdb</code> (falling back to <code>~/.local/share/knack-elt/</code>), or wherever <code>--db-path</code> points. Deliberately not working-directory relative — one app, one warehouse, wherever the command runs	nothing — this is the zero-setup path, so a fresh install can load a Knack app straight away
<code>motherduck</code>	<code>md:///knack_{stable_app_id}_data</code>	<code>motherduck_api_key</code> in the environment or <code>.env</code>

Either way the run sets `load.workers = 3` and `truncate_staging_dataset = True`, reconciles confirmed-empty and removed objects, and writes `_load_info`, `_trace`, `_kn_object_catalog` and `_kn_field_catalog` beside the data. Last, if a `{stable_app_id}_labels` schema already exists, it reports any label drift since those views were last built — read-only, and never able to change the run's exit code. See [the label views section](#) above for what builds and rebuilds that schema.

The application id and REST API key are threaded from the CLI into the record client, so `--app-id` alone fully determines which app is read — metadata and records cannot disagree.

4. SCD2 row lifecycle

Every table is loaded with `write_disposition={"disposition": "merge", "strategy": "scd2"}`, so the warehouse is **append-and-retire, not overwrite**. dlt adds `_dlt_valid_from` and `_dlt_valid_to` to every row. A record's history is therefore several rows sharing one `record_id`.



Because of that last case, two different flags are needed, and conflating them is the most common way to get a wrong answer out of this stack:

```
-- one row per record, live or deleted, plus the live/deleted flag
with flagged as (
  select
    *,
    row_number() over (partition by record_id order by _dlt_valid_from desc) = 1
      as latest_version,
    _dlt_valid_to is null as is_live_in_knack
  from acme_ops.some_table      -- the raw, dlt-owned schema
)
select * from flagged where latest_version
```

Question	Filter
Current state of everything still in Knack	latest_version and is_live_in_knack
Latest known state of every record ever seen, including deleted	latest_version
What did this record look like on a given date	_dlt_valid_from <= d and (_dlt_valid_to is null or _dlt_valid_to > d)
What has been deleted from Knack	latest_version and not is_live_in_knack

Two traps worth stating explicitly:

- **Partition by record_id , not by a column that looks like an id.** record_id is the canonical Knack row id, taken from the API payload and populated on every row. A field_N column may hold the app's own numbering or Knack's auto-added Record ID field, but neither is the merge key. Partitioning on either collapses rows into wrong groups and corrupts the flags.
- **Aggregate without latest_version and you double-count.** Every historical version of a row is still a row. A SUM() over the raw table sums every version of every record.

Deleted-in-Knack is not the same as *archived by policy* — distinguishing an intentional retention prune from incidental deletion needs a separate manifest of what the pruning process removed; SCD2 alone cannot tell them apart.

Configuration

pydantic-settings , loaded from environment or .env (src/knack_elt/config.py):

Variable	Used for
KNACK_APP_ID	Knack application id — also the default for <code>--app-id</code>
KNACK_API_KEY	Knack REST API key (sent as <code>X-Knack-REST-API-Key</code>)
motherduck_api_key	MotherDuck token, interpolated into the <code>md:///</code> connection string

Note that dbt-duckdb reads the MotherDuck token from `motherduck_token` , not `motherduck_api_key` — the two layers disagree on the variable name, so a deployment running both needs to set or shim both.